

1. Plant biodiversity and coffee yield in Ethiopia's coffee agroforests



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Study of the relationship between biodiversity and coffee yield for sites in Ethiopia

- Tropical agroforests (where crops are grown under a cover of shade trees) provide production for farmers, but are also refuges for biodiversity
- Trade-off between managing for increased yield or biodiversity

Data:

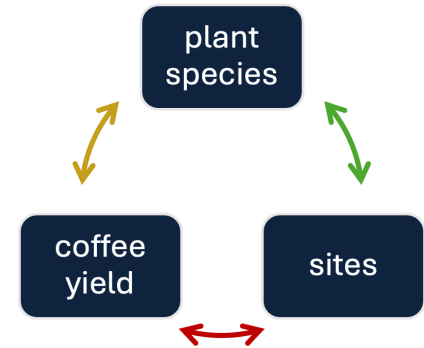
- Species richness and community composition of 407 woody plants (trees, shrubs and lianas), herbaceous plants (on the ground and epiphytic) and bryophytes (mosses and liverworts) at 60 coffee sites
- Measures of coffee management-related variables and assessment of coffee yield for 3 consecutive years at each site
- 16 coffee shrubs are considered in each site

Source: Zewdie, B., Tack, A.J., Ayalew, B., Wondafrash, M., Nemomissa, S., & Hylander, K. (2022). Plant biodiversity declines with increasing coffee yield in Ethiopia's coffee agroforests. Journal of Applied Ecology, 59, 1198–1208. <https://doi.org/10.1111/1365-2664.14130>

Data available from https://gitlab.kuleuven.be/aida-lab/teaching/g0r72a_data_ines_2526

DATA FILES

- **species x site**
 - abundance of herbaceous plant species
 - abundance of woody plant species
 - occurrence frequency of bryophyte species
 - species composition (presence/absence)
- **site x species richness**
 - species richness per species group and in total
- **species x yield**
 - number of sites with the species and average coffee yield (including to which species group each species belongs)
- **site x yield**
 - clean coffee yield for 3 years and their mean
- **site x management-related proxies**
 - coffee structure index, coffee density and coffee dominance
- **shrubs x index variables**
 - 7 measured variables and cluster index for each shrub (including to which site each shrub belongs)



PERSONAE



Hana Abebe is a coffee producer managing a coffee agroforest and is primarily interested in increasing yield across her sites. She wants to understand which site conditions, management practices, and biodiversity characteristics (including plant species richness and composition) are associated with higher coffee production, and whether there are practical ways to improve yield without introducing unnecessary complexity. She needs clear visual evidence that helps her compare sites, identify strong and weak performers, and recognise which factors appear most relevant for production.



Sofia Almeida works as an activist for an international organisation dedicated to biodiversity conservation. She is concerned that pressure to increase coffee yield may reduce species richness, alter community composition, and erode the ecological value of these agroforests. She wants to understand where biodiversity is highest, how different species groups respond to management intensity, and whether higher-yield sites systematically come at a cost to conservation. She needs visualisations that make biodiversity patterns visible and help her argue for management choices that better protect ecological integrity.



Elena Novak is an agroecology scientist planning a new field study on biodiversity in coffee-producing sites. She wants to understand which yield variable is most appropriate to measure in future studies, how strongly the available yield measures agree with one another, and which environmental or management variables should be prioritised alongside them. She is looking for interpretable visual evidence that helps her assess the structure of the dataset, compare the usefulness of the different yield indicators, and make informed methodological choices for future sampling and analysis.